EPSR-test-ex29 makemole: interactive answers, rotation options and column-map format

Program(s): makemole | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises the less common paths of `makemole` [1]: missing bond lengths, angles, potential parameters and density asked on standard input, extra distance constraints, whole-molecule and intersecting rotational groups, massless charge sites, and the original column-map `.mol` format that draws the molecule as a grid of labels and bonds.

Code exercised

Fortran (original, fortran_src/):

- `makemole.f` -> `ato_manipulation_rev4.f`: `makemole` (`.gmol` and column-map readers, stdin prompts, `extra`, `wholerot`, `introt`, `qradius`, `changelabel`, `pseudofac`, `.atm` map)
- `ato_manipulation_rev4.f`: `get_rotgrp`, `get_whole_rotgrp`, `ftype`
- `ato_read_and_save.f`: `moloutc`

Rust (port, rust/epsr-core/src/):

- `tools/makemole.rs`: `run`, `get_rotgrp`, `get_whole_rotgrp`, `read_char_int` (column-map `map` entries)
- `ato.rs`: `ftype`, `moloutc`

Physical system

Three hand-made molecules. `Chain`: an ethanol-like CH3-CH2-OH chain (9 atoms) with two extra H...H distances, rotations about C1-C2 and C2-O3, a whole-molecule group and an intersection group, the hydroxyl O relabelled `OH`; the density 0.05 atoms/Å³ is typed on stdin (box 5.646 Å) and the invalid temperature -5 is ignored (300 K). `Tip4p`: a 4-site water with the TIP4P geometry and charges (O-H 0.9572 Å, H-O-H 104.52°, massless site M 0.15 Å from O carrying -1.04 e, H +0.52 e) [2]; M is a Q atom excluded from the atom count, so 3 atoms at 0.1 atoms/Å³ give a 3.107 Å box at 298 K. `Grid`: ethane (2 C + 6 H) drawn on a column map with `nspace 2`, a repeat count `2H` and relative/absolute `map` entries; 8 atoms at 0.1 atoms/Å³ give a 4.309 Å box.

Inputs

`Chain.mol`, `Tip4p.mol` and `Grid.mol` written for this case; answers in `chain.stdin` (two bond lengths, four angles, the density, and the potential parameters of the labels `OH` and `HO`, which have no `potential` line) and `grid.stdin` (one missing angle, 95°).

How the accuracy was verified

`commands.txt` (3 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

One pass of three `makemole` calls; seven compared files: `Chain.ato`, `Chain.mol`, `Tip4p.ato`, `Tip4p.mol`, `Grid.ato`, `Grid.mol` and the atom-number map `Grid.atm`, which records the numbering of the grid entries.

Notes

Reproduced Fortran quirk: the `ROT` label of whole-molecule groups is lost (the Fortran writes the zero bytes of a newly allocated element). `pseudofac 0` is replaced by 1.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] W. L. Jorgensen, J. Chandrasekhar, J. D. Madura, R. W. Impey and M. L. Klein, "Comparison of simple potential functions for simulating liquid water", J. Chem. Phys. 79, 926-935 (1983).

[3] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.